SIMATS ENGINEERING
(SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis
Course Code: CSA1017 | Course Title: Software Engineering

Project Title	AI-Powered Smart Volunteer and NGO Resource Management Platform
Course Name	Software Engineering
Course Code	CSA1017
Faculty	R. Latha
Student Name	R. Indhu
Register Number	192524332
Department	AI & DS
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1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document describes the functional and non-functional requirements for the proposed "AI-Powered Smart Volunteer and NGO Resource Management Platform." The purpose of this document is to provide a complete, unambiguous, and verifiable description of the system to be developed, so that it may serve as a common reference point for students, evaluators, developers, project stakeholders, and Non-Governmental Organizations (NGOs) who intend to adopt the platform. This document follows the scenario-based requirement analysis approach and is prepared as part of the CO2 Assessment Tool for the course Software Engineering (CSA10).

1.2 Document Conventions

Requirement identifiers follow the pattern FR-XX for Functional Requirements and NFR-XX for Non-Functional Requirements. Priority levels are categorized as High, Medium, and Low. Use case identifiers follow the pattern UC-XX. All mandatory requirements are described using the term "shall," while desirable or optional features are described using the term "should."

1.3 Intended Audience and Reading Suggestions

- NGO administrators and project coordinators who will configure and operate the platform.
- Volunteers and donors who will use the mobile and web interfaces.
- Software developers, architects, and QA engineers responsible for building and testing the system.
- Academic evaluators assessing the requirement analysis and design competency of the student.
- Funding agencies and regulatory bodies interested in transparency and impact reporting.

1.4 Scope of the System

The proposed system is a cloud-based, AI-enabled platform that digitizes and automates the end-to-end lifecycle of volunteer engagement and NGO resource management. The scope of the system includes:

- Digital registration and profiling of volunteers, including skills, interests, availability, and location.
- AI-based matching of volunteers to suitable NGO tasks and projects using machine learning models.

The system does not include payroll processing for paid NGO staff, government tax-filing integration, or hardware provisioning for field operations; these are treated as out of scope for the current release and may be considered in future enhancements.

2. Scenario Study and Problem Analysis

2.1 Industry Context

Non-Governmental Organizations (NGOs) and volunteer groups play a critical role in coordinating humanitarian services, education programs, healthcare camps, disaster relief operations, and community development initiatives. Efficient volunteer management and optimal resource allocation are essential for maximizing social impact. The convergence of Artificial Intelligence, cloud computing, Geographic Information Systems (GIS), and mobile technologies now enables intelligent coordination, real-time monitoring, and data-driven decision-making for the social sector.

2.2 Problem Description

NGOs often face difficulty in managing volunteers, assigning tasks, tracking donations, and monitoring project outcomes because these activities are handled through fragmented, manual, or paper-based systems such as spreadsheets, phone calls, and disconnected messaging groups. An AI-powered NGO management platform is required to intelligently match volunteers to suitable activities, optimize resource allocation, monitor project progress, track donations transparently, and provide real-time dashboards for administrators — thereby improving transparency, volunteer engagement, and overall operational efficiency.

2.3 Business Domain

The system operates within the social welfare and non-profit management domain, spanning sub-domains of healthcare outreach, primary and adult education, disaster response logistics, and community infrastructure development. It intersects with allied domains such as financial technology (donation processing), geospatial services (volunteer-task proximity matching), and data analytics (impact reporting).

2.4 Stakeholders

Stakeholder	Role / Interest in the System
Volunteers	Register, discover, and perform suitable tasks; track personal contribution history.
NGO Administrators	Create projects, assign tasks, manage resources, and monitor overall performance.
Project Coordinators	Supervise field-level execution and communicate with assigned volunteers.
Donors	Contribute funds or materials and track how contributions are utilized.
System Administrators	Maintain platform infrastructure, manage user accounts, and ensure security.
Beneficiary Communities	Indirect stakeholders who receive the services delivered through NGO projects.
Regulatory / Funding Bodies	Require transparency reports and audit trails for compliance and grant renewal.

2.5 Existing Challenges

- Manual volunteer-task matching leads to mismatched skills and low engagement.
- Disconnected spreadsheets and messaging apps cause delayed communication and lost records.
- Absence of a real-time dashboard limits administrators' visibility into ongoing projects.
- Donation tracking is opaque, reducing donor confidence and repeat contributions.

2.6 System Objectives

1. Manage volunteer registrations digitally through a unified platform.
2. Match volunteers to suitable tasks using AI-driven recommendation algorithms.
3. Monitor NGO project progress in real time.
4. Track donations and resources with full transparency and audit trails.

5. Optimize volunteer allocation across concurrent projects and geographies.

2.7 Product Perspective

The platform is a new, self-contained, cloud-hosted product accessible through a volunteer mobile application, an NGO administrator web portal, and a donor web portal. It integrates with external systems including payment gateways for donation processing, SMS/email gateways for notifications, and mapping/GIS APIs for location-based volunteer-task optimization.

3. Functional Requirements

Functional requirements define the specific behaviors, features, and functions that the system shall provide. They are grouped below by functional module for clarity and traceability.

3.1 Volunteer Management Module

ID	Requirement Description	Priority
FR-01	The system shall allow volunteers to register using email, phone number, or social login.	High
FR-02	The system shall allow volunteers to create and update a profile containing skills, interests, availability, and location.	High
FR-03	The system shall allow volunteers to upload supporting documents (ID proof, certifications).	Medium
FR-04	The system shall allow volunteers to browse and apply for available tasks and projects.	High
FR-05	The system shall allow volunteers to track their cumulative volunteering hours and history.	Medium

3.2 AI-Based Matching Module

ID	Requirement Description	Priority
FR-07	The system shall use an AI matching engine to recommend suitable tasks to volunteers based on skills, interest, location, and availability.	High

ID	Requirement Description	Priority
FR-08	The system shall rank and suggest alternative volunteers to NGO administrators when a task remains unfilled.	Medium
FR-09	The system shall continuously refine matching accuracy using historical assignment and feedback data.	Medium
FR-10	The system shall allow administrators to override AI-suggested matches manually.	High

3.3 Project and Task Management Module

3.4 Donation and Resource Management Module

ID	Requirement Description	Priority
FR-16	The system shall allow donors to make monetary donations through an integrated payment gateway.	High
FR-17	The system shall allow donors and administrators to record in-kind (material) donations.	Medium
FR-18	The system shall maintain a resource inventory of donated and procured materials.	Medium
FR-19	The system shall allow donors to track how their donations are utilized across specific projects.	High

3.5 Reporting, Dashboard, and Notification Module

ID	Requirement Description	Priority
FR-21	The system shall provide a real-time dashboard displaying volunteer participation, project status, and donation summary.	High
FR-22	The system shall generate downloadable organizational reports (PDF/Excel) for a selected date range.	Medium
FR-23	The system shall send automated notifications (SMS, email, push) for task assignment, reminders, and status updates.	High

4. Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints that the system must satisfy in addition to its functional behavior.

ID	Category	Requirement Description
NFR-01	Performance	The system shall respond to volunteer-task match requests within 3 seconds under normal load (up to 5,000 concurrent users).
NFR-02	Security	The system shall encrypt all sensitive data (personal information, payment details) using industry-standard encryption (AES-256, TLS 1.2 or higher) both in transit and at rest.
NFR-03	Security	The system shall implement role-based access control (RBAC) to restrict functionality based on user roles.
NFR-04	Reliability	The system shall maintain 99.5% uptime on a monthly basis, excluding scheduled maintenance.
NFR-05	Usability	The system shall provide an intuitive, mobile-responsive interface usable by volunteers with minimal digital literacy, supporting at least two languages.

5. Actors and Use Cases

5.1 Actor Identification

Actor	Type	Description
Volunteer	Primary	An individual who registers, applies for tasks, and performs volunteering activities.
NGO Administrator	Primary	Manages NGO projects, resources, task assignment, and organizational reporting.
Project Coordinator	Primary	Supervises field execution of specific projects and communicates with volunteers.
Donor	Primary	Contributes funds or materials and monitors utilization of donations.

Actor	Type	Description
System Administrator	Primary	Manages user accounts, roles, permissions, and overall platform configuration.
AI Matching Engine	Secondary	Automated component that analyzes profiles and recommends volunteer-task matches.

5.2 Use Case Diagram

The following use case diagram illustrates the interactions between the identified actors and the major use cases supported by the AI-Powered Smart Volunteer and NGO Resource Management Platform.

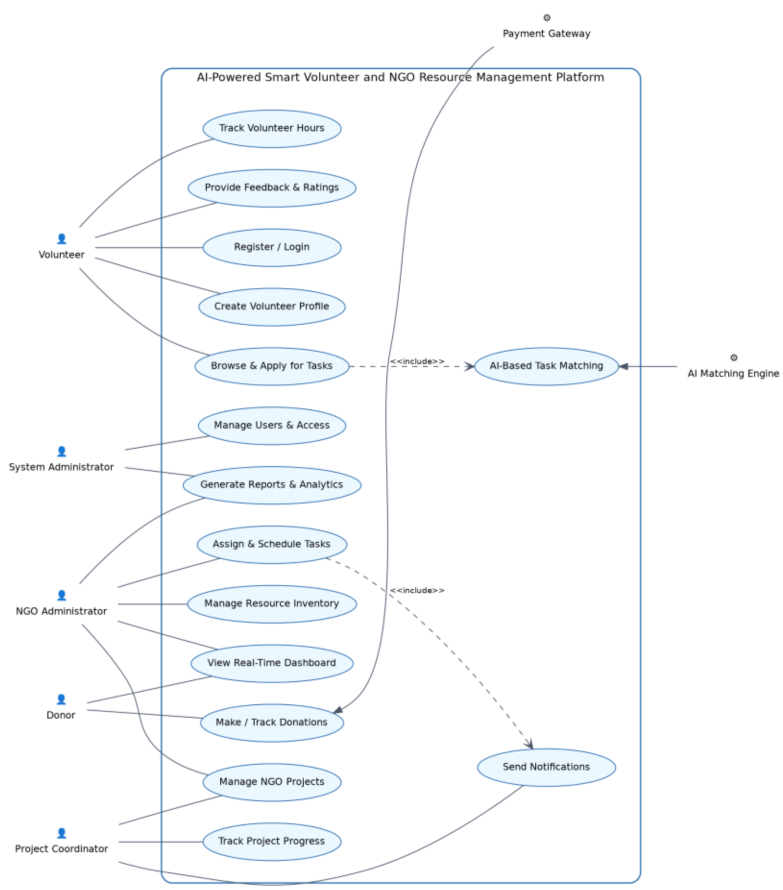

Figure 1: Use Case Diagram — AI-Powered Smart Volunteer and NGO Resource Management Platform

5.3 Use Case Descriptions

Detailed descriptions of the five most critical use cases are provided below, following standard use case specification format.

UC-01: AI-Based Task Matching

Use Case ID	UC-01 — AI-Based Task Matching
Actors	Volunteer, AI Matching Engine
Precondition	Volunteer has created a complete profile with skills and availability; at least one active task exists.
Description	The system automatically recommends the most suitable tasks to a volunteer based on skill fit, location proximity, and availability.
Main Flow	1. Volunteer logs into the mobile application. 2. System retrieves the volunteer profile and active task list. 3. AI Matching Engine computes compatibility scores between the volunteer and available tasks. 4. System displays a ranked list of recommended tasks to the volunteer. 5. Volunteer selects and applies for a recommended task.
Postcondition	Volunteer is matched with a suitable task and notified of the application status.
Exception Flow	If no suitable match is found, the system suggests the closest alternative tasks and notifies the NGO administrator of an unfilled requirement.

UC-02: Assign and Schedule Tasks

Use Case ID	UC-02 — Assign and Schedule Tasks
Actors	NGO Administrator, Project Coordinator
Precondition	A project and its associated tasks have been created in the system.
Description	The NGO administrator assigns matched volunteers to specific tasks and schedules the activity.
Main Flow	1. Administrator reviews AI-recommended volunteers for a task.

	2. Administrator selects a volunteer and confirms the assignment. 3. System schedules the task with date, time, and location. 4. System sends an automated notification to the assigned volunteer. 5. Volunteer accepts or declines the assignment.

UC-03: Track Donations

Use Case ID	UC-03 — Track Donations
Actors	Donor, Payment Gateway, NGO Administrator
Precondition	Donor is registered on the platform and a donation campaign or project is active.
Description	The donor contributes funds or materials and monitors how the contribution is utilized.
Main Flow	1. Donor selects a project or general fund to contribute to. 2. Donor enters the donation amount and payment details. 3. System forwards the transaction to the payment gateway for processing. 4. System records the confirmed transaction and issues a digital receipt. 5. Donor views utilization updates linked to their contribution on the dashboard.

UC-04: Monitor Project Progress

Use Case ID	UC-04 — Monitor Project Progress
Actors	Project Coordinator, NGO Administrator
Precondition	A project has been created with defined milestones and assigned tasks.

Description	The coordinator updates and the administrator monitors real-time progress of an ongoing project.
Main Flow	1. Coordinator logs completed tasks and milestone status from the field. 2. System recalculates overall project completion percentage. 3. System updates the real-time dashboard visible to administrators and donors. 4. Administrator reviews progress and reallocates resources if delays are detected.
Postcondition	Project progress is accurately reflected across all stakeholder dashboards.
Exception Flow	If a milestone is significantly delayed, the system flags the project as “At Risk” and alerts the administrator.

UC-05: Generate Reports and Analytics

Use Case ID	UC-05 — Generate Reports and Analytics
Actors	NGO Administrator, System Administrator
Precondition	Sufficient historical data (volunteers, tasks, donations) exists in the system.
Description	The administrator generates consolidated reports for internal review or external funding bodies.
Main Flow	1. Administrator selects the report type and date range. 2. System aggregates data from volunteer, project, and donation modules. 3. System generates a formatted report (PDF/Excel) with charts and summaries. 4. Administrator downloads or shares the report with stakeholders.

Postcondition	A consolidated, exportable report is generated reflecting accurate organizational performance.
Exception Flow	If insufficient data exists for the selected range, the system displays a notification instead of an empty report.

6. System Features

6.1 Intelligent Volunteer-Task Matching

Description: Uses machine learning to analyze volunteer skills, past performance, location, and availability against task requirements to generate ranked recommendations.

Priority: High. Stimulus/Response: When a volunteer opens the “Find Tasks” screen, the system shall display a ranked list of matching tasks within 3 seconds.

6.2 Real-Time Administrative Dashboard

Description: Provides NGO administrators with a consolidated, visual overview of active volunteers, ongoing projects, donation inflow, and task completion rates.

Priority: High. Stimulus/Response: Dashboard metrics shall refresh automatically at least every 5 minutes or on manual refresh.

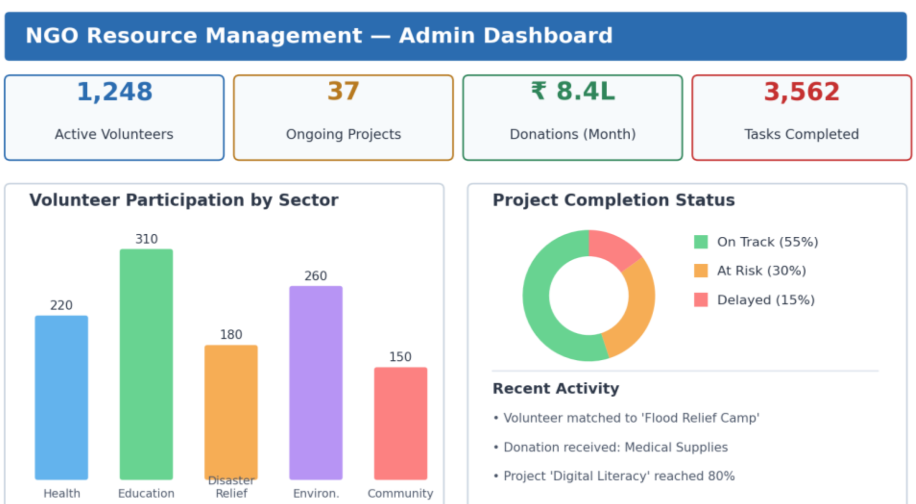

Figure 2: Illustrative Mockup of the NGO Administrator Real-Time Dashboard

6.3 Transparent Donation and Resource Tracking

Description: Records every monetary and in-kind donation against a specific project and provides donors with utilization visibility, improving trust and repeat contributions.

Priority: High. Stimulus/Response: Upon successful payment confirmation, the system shall generate a digital receipt and update the donor's contribution ledger immediately.

6.4 Multi-Channel Notification Engine

Description: Automatically informs volunteers, coordinators, and donors of relevant events such as task assignment, schedule changes, milestone completion, and donation confirmations through SMS, email, and push notifications.

Priority: Medium. Stimulus/Response: The system shall dispatch a notification within 60 seconds of the triggering event.

6.5 Role-Based Access and User Management

Description: Allows the system administrator to define and manage distinct roles (volunteer, coordinator, NGO admin, donor, system admin) with appropriately scoped permissions.

Priority: High. Stimulus/Response: Any unauthorized access attempt shall be blocked and logged for audit review.

7. Data Requirements

The system shall persist and manage the following core data entities. Field lists are representative and not exhaustive.

Entity	Key Attributes	Description
Volunteer	VolunteerID, Name, Contact, Skills, Location, Availability, HoursLogged	Stores volunteer identity, capability, and participation history.
NGO	NGOID, Name, RegistrationNumber, Domain, ContactPerson	Represents a registered NGO organization operating on the platform.
Project	ProjectID, NGOID, Title, Domain, StartDate, EndDate, Status	Represents a specific initiative undertaken by an NGO.

Entity	Key Attributes	Description
Task	TaskID, ProjectID, RequiredSkills, Location, Schedule, Status	Represents an individual assignable unit of work within a project.
Assignment	AssignmentID, TaskID, VolunteerID, MatchScore, Status	Links a volunteer to a task with the AI-generated match score.
Donation	DonationID, DonorID, ProjectID, Amount/Item, Date, PaymentStatus	Records monetary or in-kind contributions linked to a project.
Donor	DonorID, Name, Contact, DonationHistory	Stores donor identity and cumulative contribution record.
Notification	NotificationID, UserID, Type, Message, Timestamp, Status	Tracks outbound communication events sent to users.
User Account	UserID, Role, Credentials, AccessLevel	Manages authentication and role-based authorization data.

7.1 Data Retention and Integrity

- Donation and financial transaction records shall be retained for a minimum of five years to satisfy audit and regulatory requirements.
- Referential integrity shall be enforced between Volunteer, Task, and Assignment entities to prevent orphaned records.
- Personal data fields shall be subject to access control and shall be anonymized in analytical reports wherever individual identification is not required.

8. External Interface Requirements

8.1 User Interfaces

- A responsive volunteer mobile application (Android and iOS) for registration, task discovery, and progress tracking.
- A web-based NGO administrator portal for project creation, task assignment, resource management, and reporting.

- A donor web portal for making contributions and viewing utilization dashboards.
- All interfaces shall follow consistent visual design guidelines and support accessibility best practices (readable font sizes, color contrast, screen-reader compatibility).

8.2 Hardware Interfaces

- Volunteer mobile devices shall support GPS for location-based task matching, where the volunteer opts in.
- Server infrastructure shall run on cloud-hosted virtual machines or containers with elastic scaling capability.

8.3 Software Interfaces

Interface	Purpose
Payment Gateway API	Processes monetary donations securely and returns transaction confirmation.
SMS / Email Gateway API	Delivers transactional notifications to volunteers, donors, and administrators.
Maps / GIS API	Supplies geocoding and distance-calculation data for location-based volunteer-task matching.
Cloud Database Service	Provides managed, scalable storage for structured application data.
Authentication Provider (OAuth2 / JWT)	Handles secure login and session management across web and mobile clients.

8.4 Communication Interfaces

- All client-server communication shall occur over HTTPS using RESTful APIs secured with TLS 1.2 or higher.
- Real-time dashboard updates may use WebSocket connections for low-latency data refresh.

9. System Architecture

The platform follows a layered, service-oriented architecture comprising a presentation layer, an application/API layer, an AI/intelligence layer, a data layer, and integrations with external systems. This separation of concerns supports the scalability, maintainability, and portability requirements defined in Section 4.

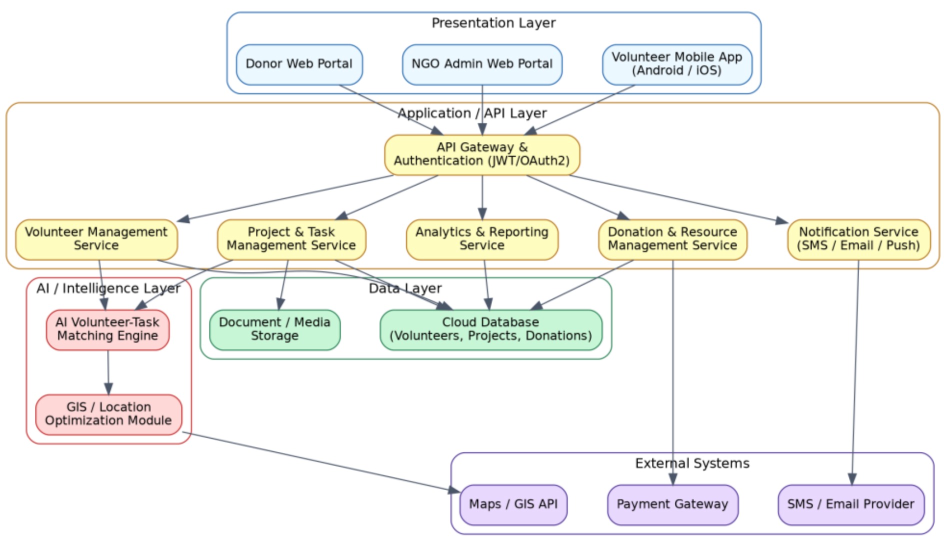

Figure 3: High-Level System Architecture of the AI-Powered Smart Volunteer and NGO Resource Management Platform

9.1 Layer Description

Layer	Responsibility
Presentation Layer	Volunteer mobile app, NGO admin web portal, and donor web portal deliver the user-facing experience.
Application / API Layer	API gateway handles authentication and routes requests to dedicated microservices for volunteers, projects, donations, notifications, and reporting.
AI / Intelligence Layer	Hosts the AI matching engine and GIS-based location optimization module used for volunteer-task recommendations.

10. Process Flow — AI-Based Task Matching and Assignment

The flowchart below illustrates the end-to-end process flow from volunteer registration through AI-based matching, task assignment, execution, and feedback collection.

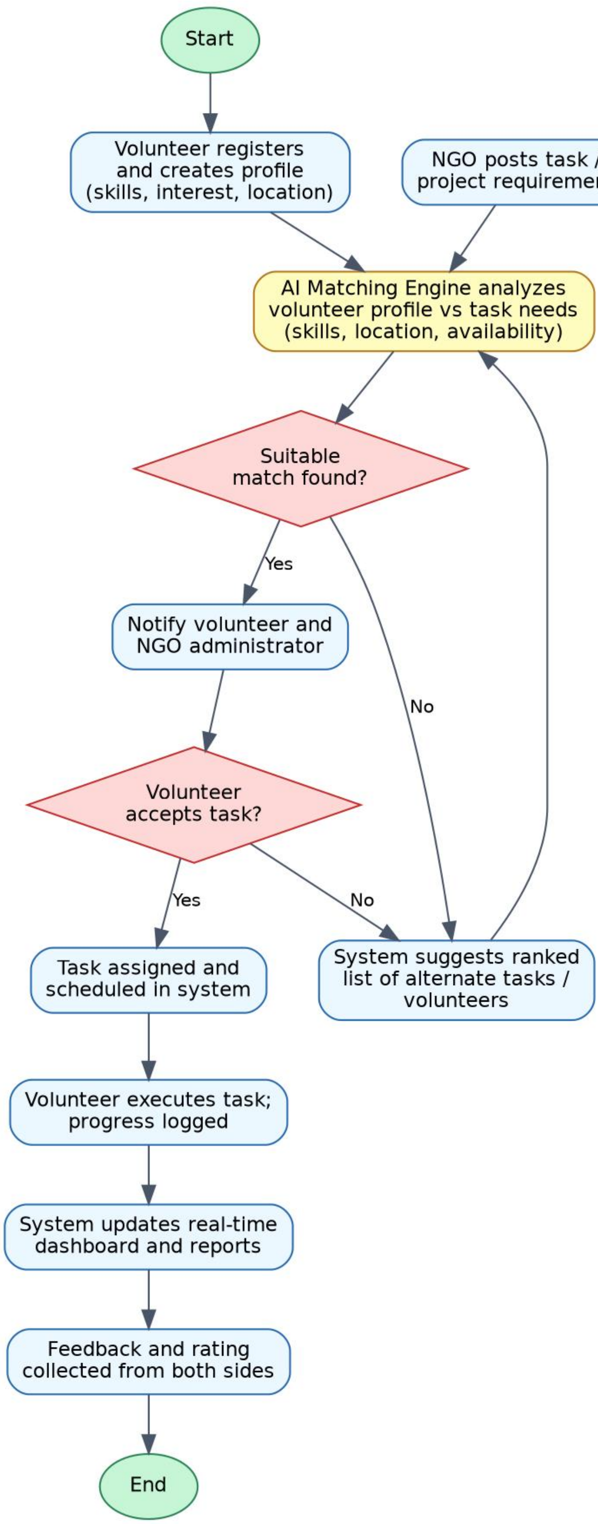

Figure 4: Process Flowchart — Volunteer-Task Matching and Assignment Workflow

11. Assumptions and Constraints

11.1 Assumptions

- Volunteers and NGO staff have access to smartphones or computers with internet connectivity.
- NGOs will provide accurate and timely information about projects, tasks, and resource requirements.
- Volunteers will maintain accurate and updated profile information for effective AI matching.
- Third-party services (payment gateway, SMS/email provider, maps API) will maintain agreed service-level availability.
- Users possess a basic level of digital literacy sufficient to operate mobile and web applications.

11.2 Constraints

Type	Constraint Description
Technical	The system must integrate with existing third-party payment and communication gateways using their published APIs and rate limits.
Operational	NGO administrators require basic training to effectively use the AI-assisted assignment override features.
Economic	Cloud infrastructure and third-party API usage costs must remain within the sponsoring NGO's operating budget.
Legal / Regulatory	The platform must comply with applicable data protection and financial transaction regulations in each operating region.
Time	The initial release must be delivered within the academic project timeline / sprint schedule defined for this assessment.
Connectivity	Field volunteers in low-connectivity areas may experience delayed synchronization of task status updates.

12. Requirement Validation — Completeness and Consistency

12.1 Stakeholder Coverage Check

Each identified stakeholder group has at least one corresponding functional requirement and use case, confirming that no primary stakeholder need has been overlooked.

Stakeholder	Addressed By
Volunteer	FR-01 – FR-06, UC-01, UC-02
NGO Administrator	FR-11 – FR-15, FR-21 – FR-24, UC-02, UC-05
Project Coordinator	FR-13, FR-14, UC-02, UC-04
Donor	FR-16 – FR-20, UC-03
System Administrator	FR-24, UC-05

12.2 Ambiguity, Redundancy, and Consistency Review

- Ambiguity: All requirements were reviewed to remove vague terms; quantifiable criteria (e.g., “within 3 seconds,” “99.5% uptime”) were added wherever measurable performance was implied.
- Redundancy: Functional requirements were grouped by module and cross-checked to ensure no duplicate requirement IDs describe the same behavior.
- Inconsistency: Terminology was standardized across the document (e.g., “task,” “project,” and “assignment” are used consistently rather than interchangeably) to avoid conflicting interpretations.
- Missing Requirements: A gap analysis against the stated objectives confirmed that reporting (FR-21, FR-22) and access control (NFR-03, FR-24) were added to fully cover transparency and security objectives that were implicit in the scenario.

12.3 Suggested Improvements

1. Introduce a formal requirements traceability matrix linking each FR/NFR to specific test cases during the design phase.
2. Conduct stakeholder walkthroughs with actual NGO administrators to validate assumptions about digital literacy and connectivity.
3. Define explicit data retention and deletion policies in collaboration with legal counsel prior to development.
4. Establish measurable acceptance criteria for the AI matching engine's recommendation accuracy (e.g., minimum precision threshold) before release.

13. Expected Outcomes

Successful implementation of the AI-Powered Smart Volunteer and NGO Resource Management Platform is expected to deliver the following measurable outcomes for NGOs, volunteers, and donors:

1. Improved volunteer participation through simplified discovery and AI-driven task matching.
2. Efficient resource allocation across concurrent projects and geographies.
3. Better project management through real-time progress monitoring and milestone tracking.
4. Enhanced organizational transparency in resource and donation utilization.
5. Increased donor confidence resulting from traceable, receipted contributions.
6. Reduced administrative workload through automation of matching, scheduling, and reporting.
7. Improved community services driven by better-coordinated volunteer efforts.
8. Greater overall social impact, supported by consolidated analytics and reporting for funding bodies.

14. Conclusion

This Software Requirement Specification presents a comprehensive, scenario-based requirement analysis for the AI-Powered Smart Volunteer and NGO Resource Management Platform. Through systematic identification of stakeholders, functional and non-functional requirements, actors, use cases, and system architecture, this document establishes a validated foundation for subsequent design and development phases. The proposed platform directly addresses the fragmentation, inefficiency, and lack of transparency currently faced by NGOs, and is expected to significantly improve volunteer engagement, resource utilization, and social impact reporting once implemented.